Capstone Project: Enterprise Azure Infrastructure Deployment

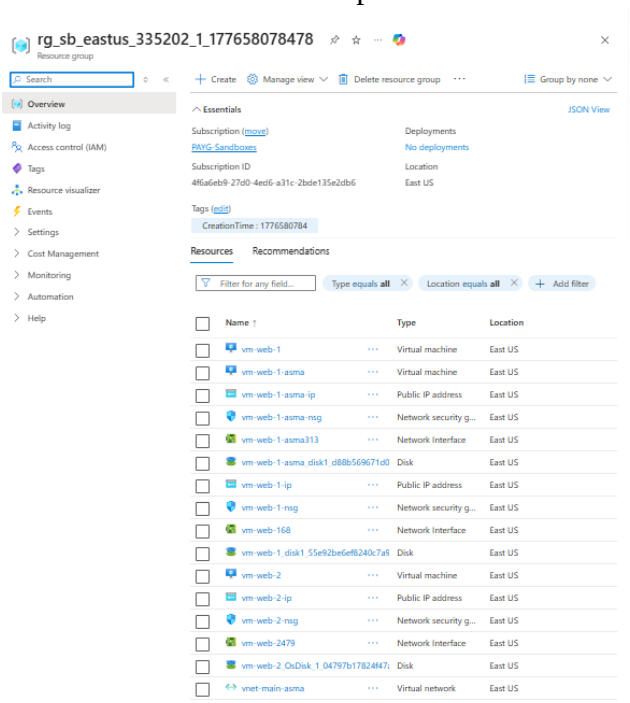
Asma Salim Khamis Al Kharusi

In this lab, I deployed a multi-tier application on Microsoft Azure including Compute, Network, Load Balancer, Database, Security, Monitoring, and Automation.

Project Requirements

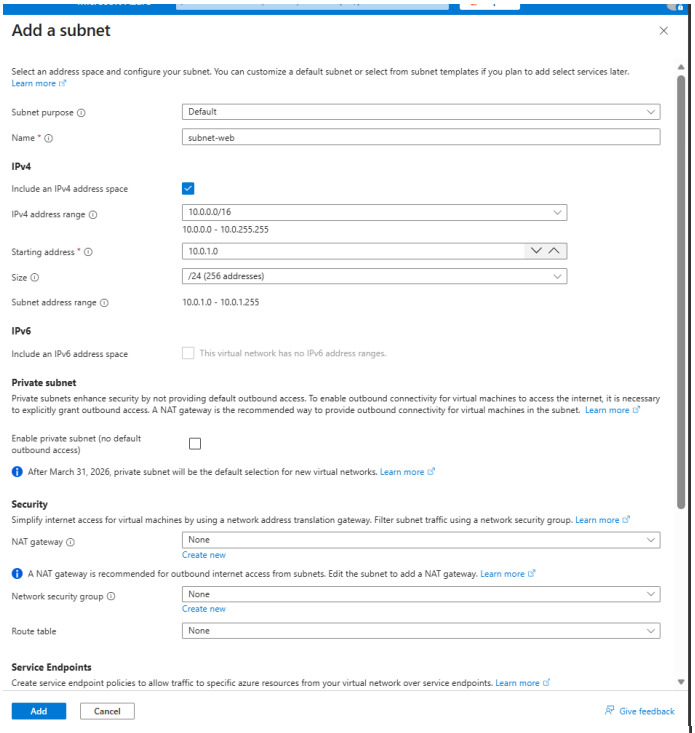
1. Resource Organization

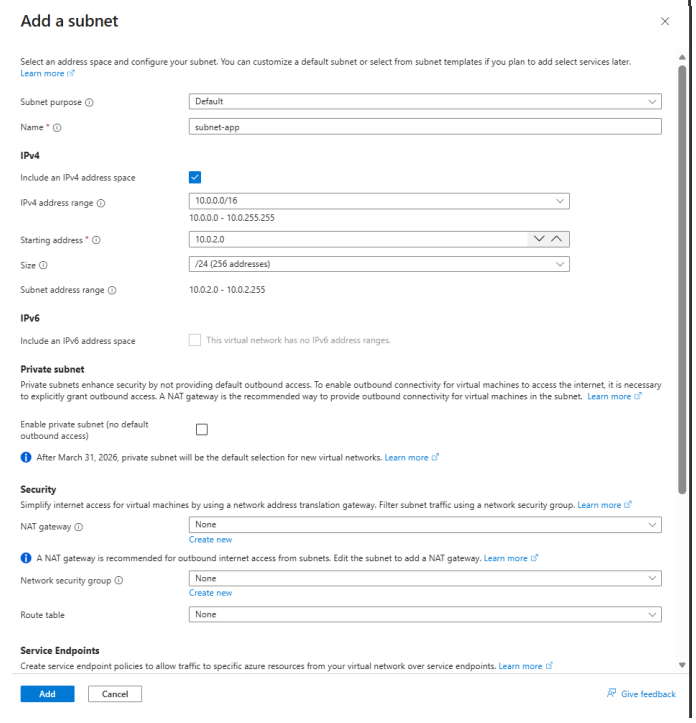
- Create a Resource Group to contain all resources



I created a Resource Group to organize and manage all Azure resources in a single container. This approach simplifies resource management, monitoring, and deployment.

2. Networking Layer





Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later.
Learn more

Subnet purpose: Default

Name *: subnet-db

IPv4

Include an IPv4 address space: ☑

IPv4 address range: 10.0.0.0/16
10.0.0.0 - 10.0.255.255

Starting address *: 10.0.3.0

Size: /24 (256 addresses)

Subnet address range: 10.0.3.0 - 10.0.3.255

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. Learn more

Enable private subnet (no default outbound access): ☐

After March 31, 2026, private subnet will be the default selection for new virtual networks. Learn more

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security group. Learn more

NAT gateway: None
Create new

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. Learn more

Network security group: None
Create new

Route table: None

Service Endpoints

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. Learn more

Add | Cancel | Give feedback

Create virtual network

Basics | Security | Address space | Tags | Review + create

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. Learn more

☐ Allocate using IP address pools. Learn more

+ Add a subnet

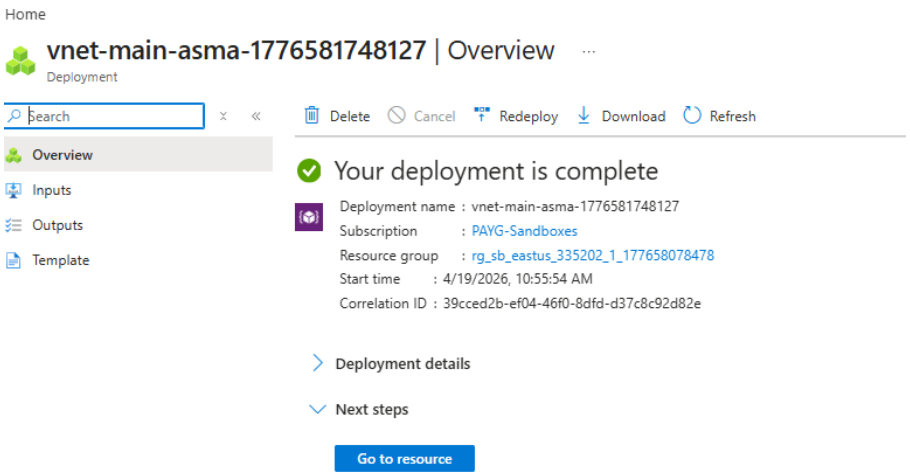
10.0.0.0/16 — Delete address space

10.0.0.0 | /16
10.0.0.0 - 10.0.255.255 | 65,536 addresses

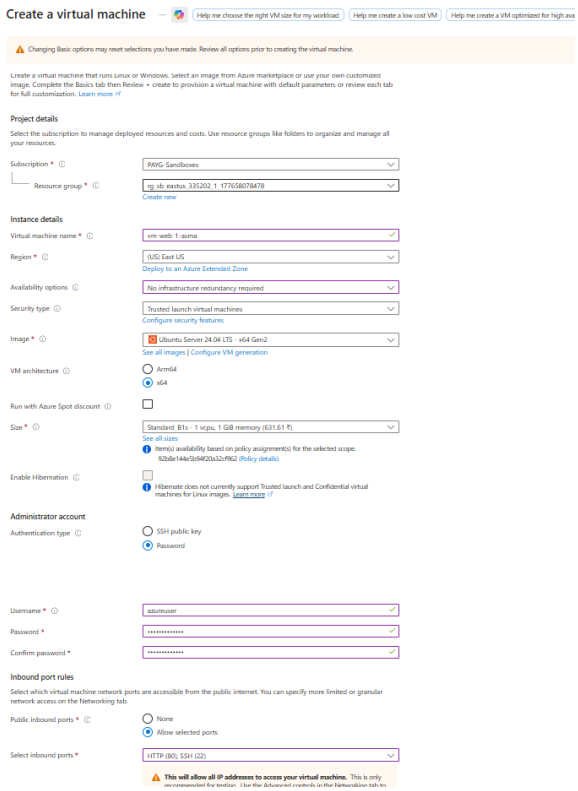
Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-
subnet-web	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-
subnet-app	10.0.2.0 - 10.0.2.255	/24 (256 addresses)	-
subnet-db	10.0.3.0 - 10.0.3.255	/24 (256 addresses)	-

Add IPv4 address space

Previous | Next | Review + create



I created a Virtual Network with three subnets (Web, App, and Database) to separate the application layers. This design improves security, scalability, and organization of resources.



Create a virtual machine

Help me choose the right VM size for my workload | Help me create a low cost VM | Help me create a VM optimized for high a

Basics | Disks | Networking | Management | Monitoring | **Advanced** | Tags | Review + create

Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions: Select an extension to install

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. Learn more

Select a VM application to install

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. Learn more about custom data for VMs

Custom data:

```
#cloud-config
package_update: true
packages:
 - nginx
runcmd:
 - echo "VM1" > /var/www/html/index.html
```

Custom data on the selected image will be processed by cloud-init.
Learn more about custom data for VMs

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. Learn more about user data for VMs

Enable user data: ☐

Performance (NVMe)

Enable capabilities to enhance the performance of your resources.

Higher remote disk storage performance with NVMe: ☐

The selected size is not supported for NVMe. See supported size families

Host

Azure Dedicated Hosts allow you to provision and manage a physical server within our data centers that are dedicated to your Azure subscription. A dedicated host gives you assurance that only VMs from your subscription are on the host, flexibility to choose VMs from your subscription that will be provisioned on the host, and the control of platform maintenance at the level of the host. Learn more

Host group: No host groups found

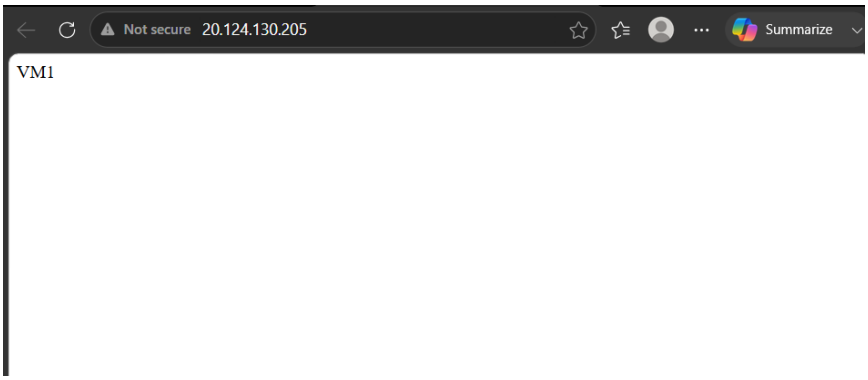
Capacity reservations

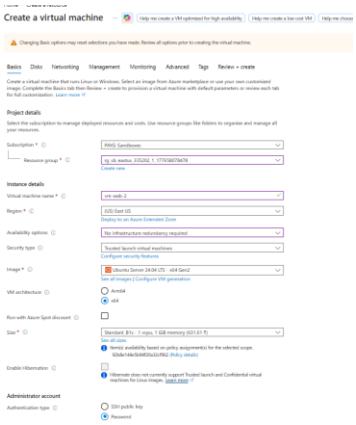
Capacity reservations allow you to reserve capacity for your virtual machine needs. You get the same SLA as normal virtual machines with the security of reserving the capacity ahead of time. Learn more

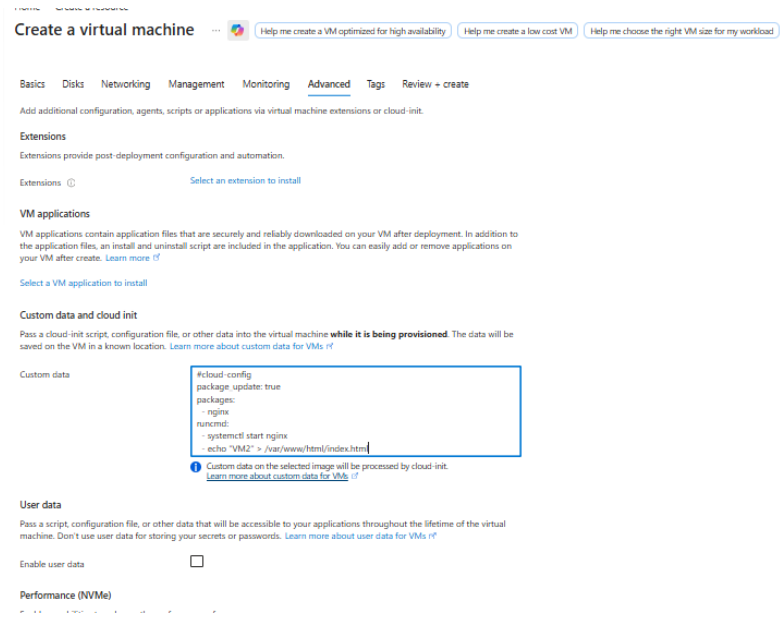
Capacity reservations:

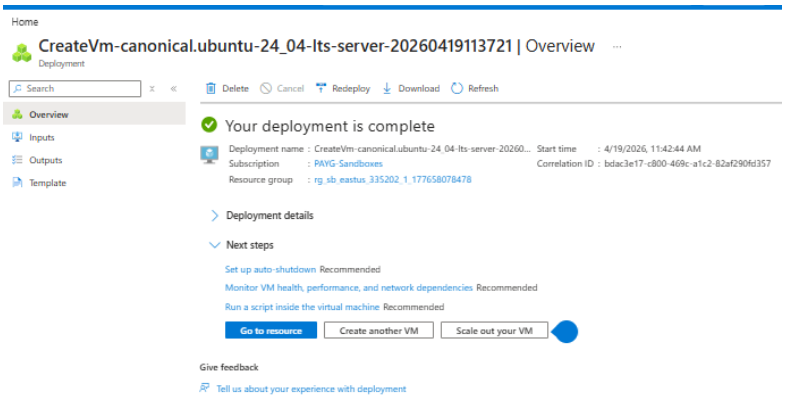
- (●) None
  Do not use reserved capacity
- ( ) Capacity reservation group
  Use reserved capacity from this subscription
- ( ) (Preview) Shared capacity reservation group

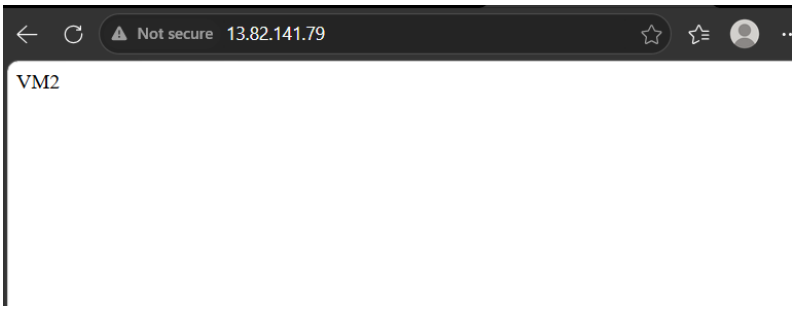
3. Compute Layer











Step: Install and Configure Web Server (Nginx)

The web server (Nginx) was successfully installed and configured on both virtual machines. This is verified by accessing the public IP address, which displays different responses (VM1 and VM2), confirming that the web servers are running correctly.

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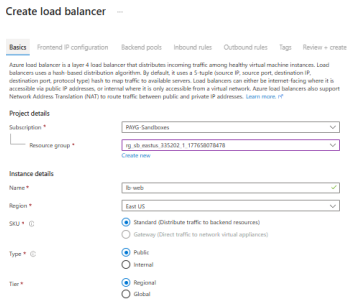
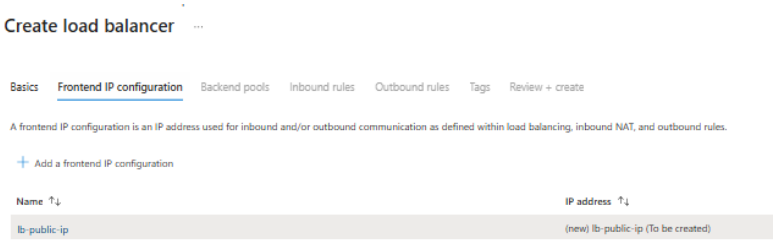
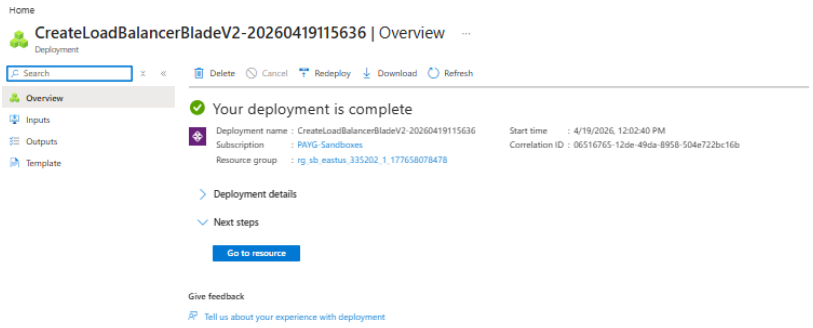
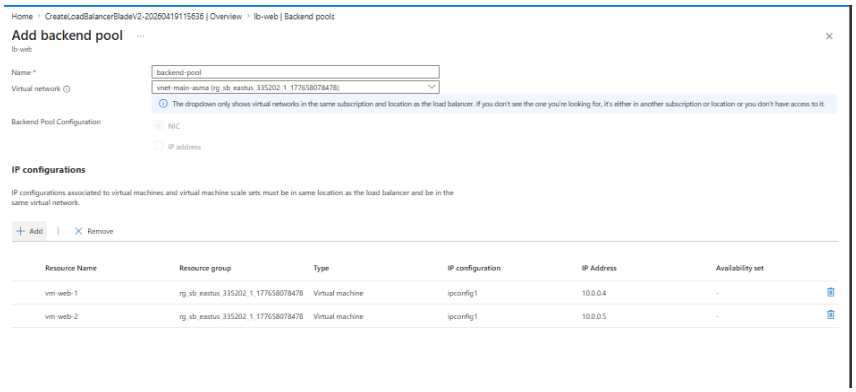
4. Load Balancing

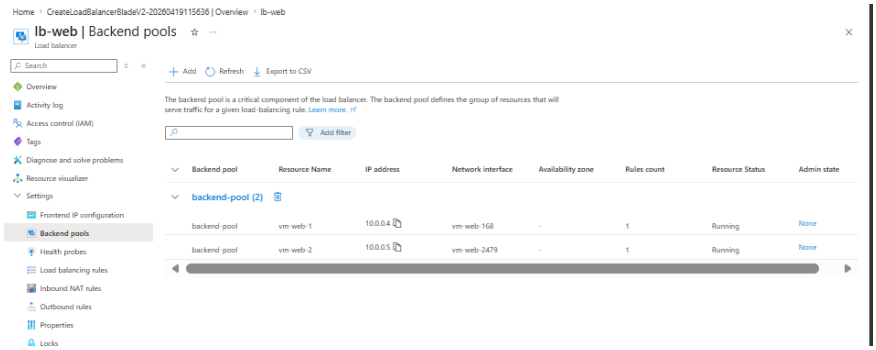




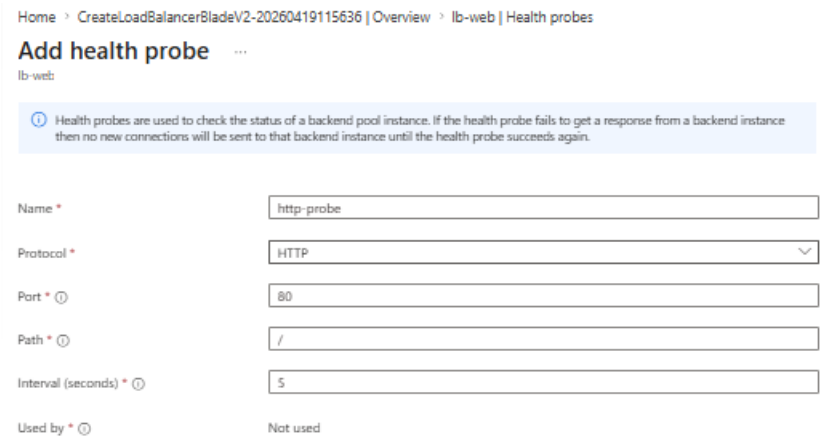


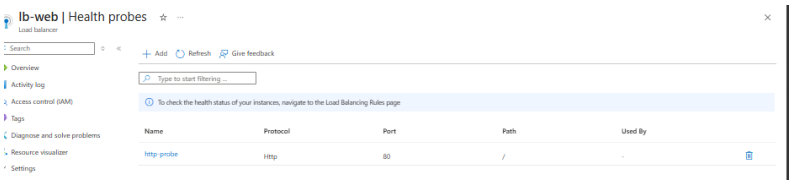
Backend Pool :



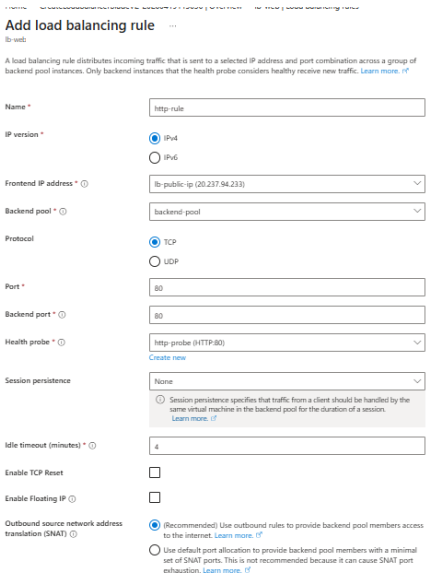


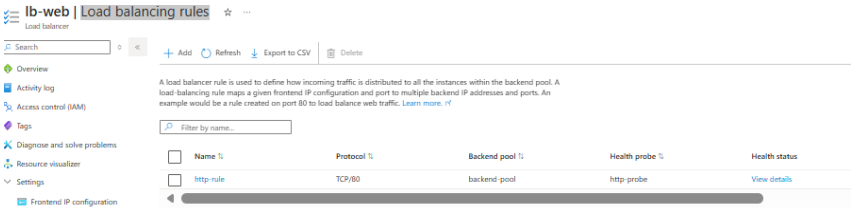
Health probes :

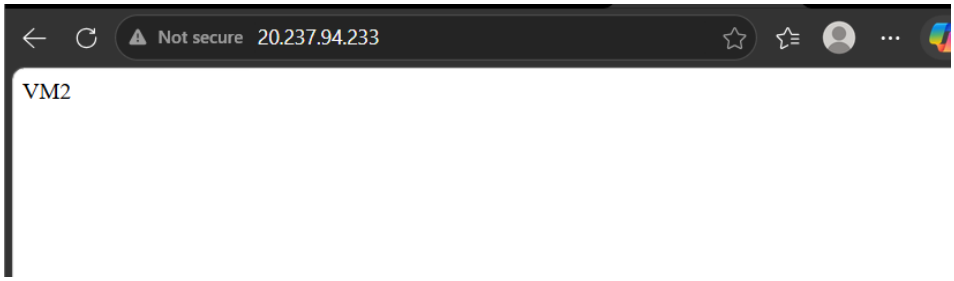


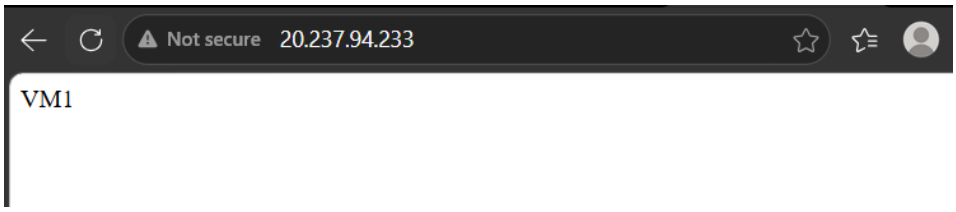


Load balancing rules :



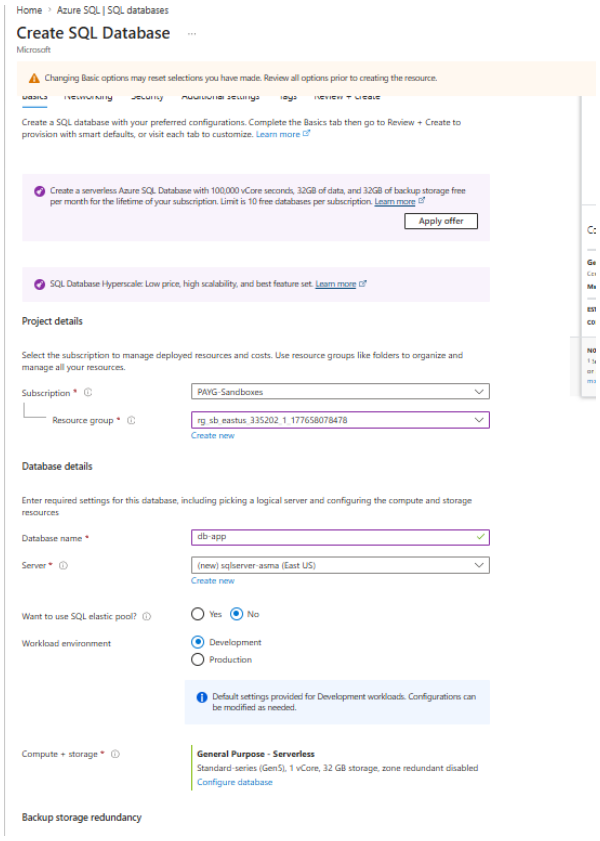


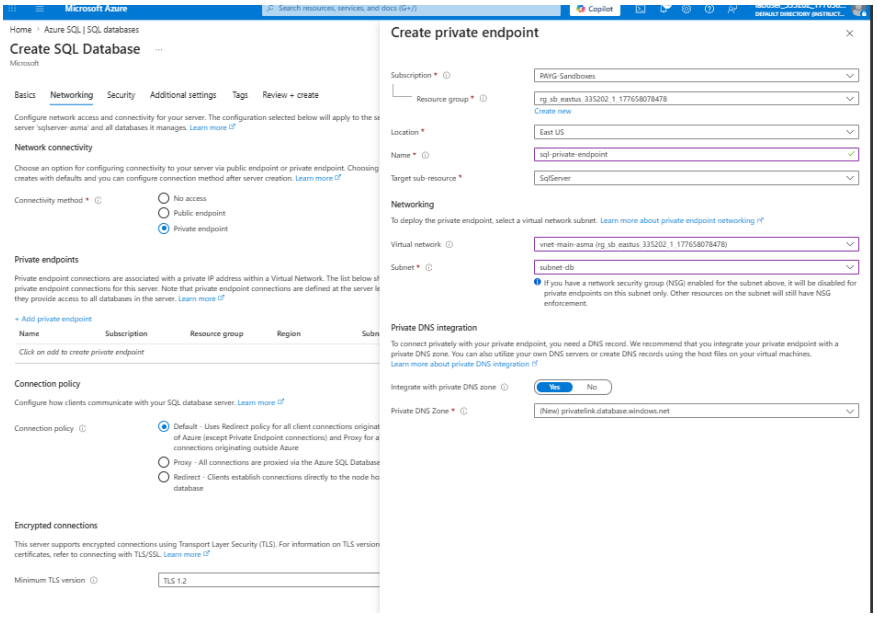


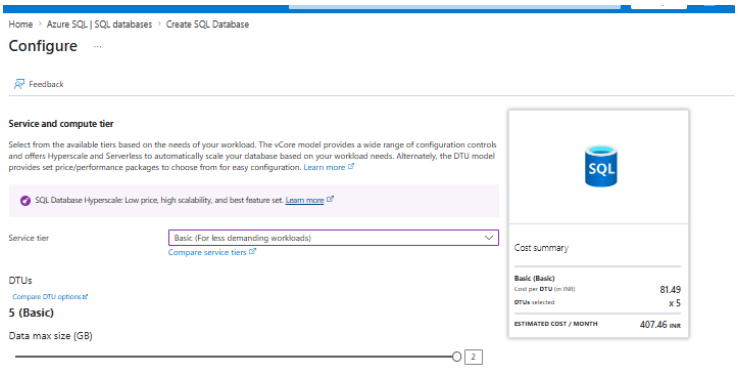


The Load Balancer was successfully tested by accessing its public IP address through a web browser. The response alternated between VM1 and VM2 upon refreshing the page, confirming that traffic is being distributed correctly across both virtual machines. This demonstrates effective load balancing and high availability of the application.

5. Data Layer


Home › Azure SQL | SQL databases
Create SQL Database
Microsoft
Changing Basic options may reset selections you have made. Review all options prior to creating the resource.
Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. Learn more
Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. Learn more
Apply offer
SQL Database Hyperscale: Low price, high scalability, and best feature set. Learn more
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
PAYG-Sandboxes
Resource group *
rg_sb_eastus_335202_1_177658078478
Create new
Database details
Enter required settings for this database, including picking a logical server and configuring the compute and storage resources
Database name *
db-app
Server *
(new) sqlserver-asma (East US)
Create new
Want to use SQL elastic pool?
Yes
No
Workload environment
Development
Production
Default settings provided for Development workloads. Configurations can be modified as needed.
Compute + storage *
General Purpose - Serverless
Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled
Configure database
Backup storage redundancy


Microsoft Azure
Home › Azure SQL | SQL databases
Create SQL Database
Microsoft
Basics
Networking
Security
Additional settings
Tags
Review + create
Network connectivity
Connectivity method *
No access
Public endpoint
Private endpoint
Private endpoints
+ Add private endpoint
Name
Subscription
Resource group
Region
Click on add to create private endpoint
Connection policy
Connection policy
Encrypted connections
Minimum TLS version
TLS 1.2
Create private endpoint
Subscription *
PAYG-Sandboxes
Resource group *
rg_sb_eastus_335202_1_177658078478
Create new
Location *
East US
Name *
sql-private-endpoint
Target sub-resource *
SqlServer
Networking
To deploy the private endpoint, select a virtual network subnet. Learn more about private endpoint networking
Virtual network
vnet-main-asma (rg_sb_eastus_335202_1_177658078478)
Subnet *
subnet-db
If you have a network security group (NSG) enabled for the subnet above, it will be disabled for private endpoints on this subnet only. Other resources on the subnet will still have NSG enforcement.
Private DNS integration
To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint with a private DNS zone. You can also utilize your own DNS servers or create DNS records using the host files on your virtual machines.
Learn more about private DNS integration
Integrate with private DNS zone
Yes
No
Private DNS Zone *
(New) privatelink.database.windows.net



Azure SQL Database was created to store application data.

Public access was disabled, and a private endpoint was used to allow access only from inside the virtual network.

This improves security.

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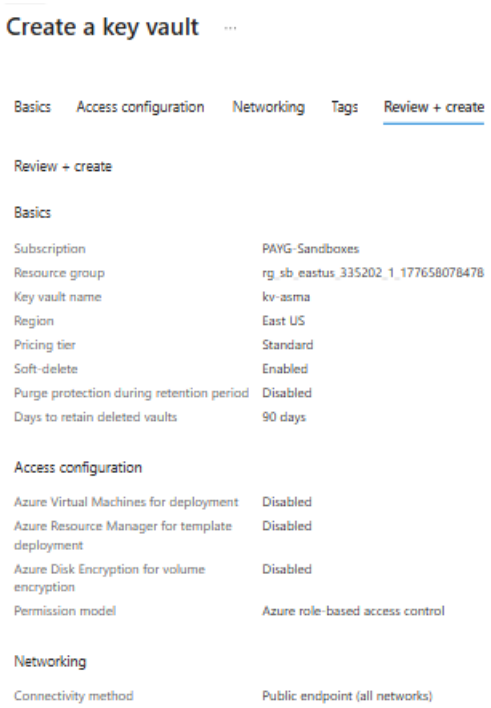
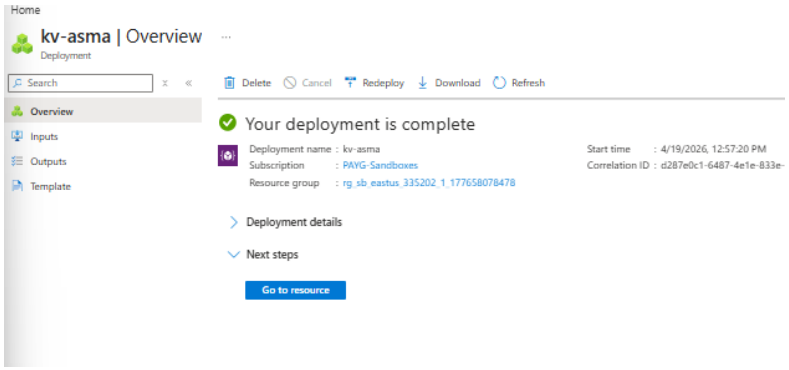
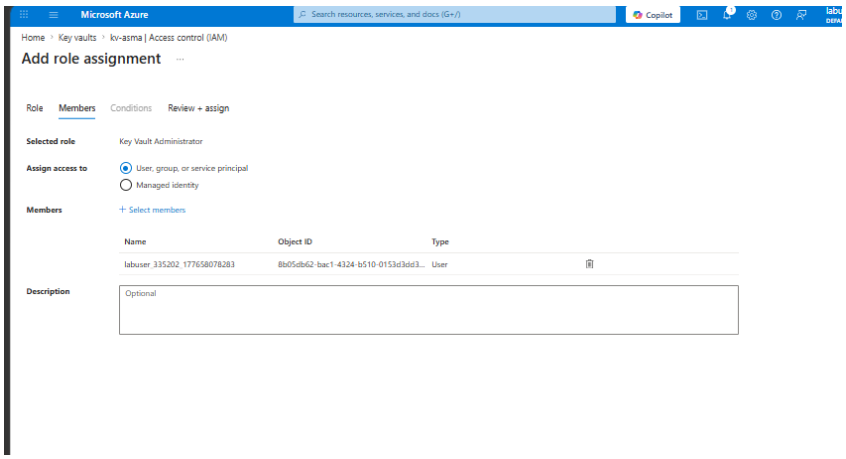
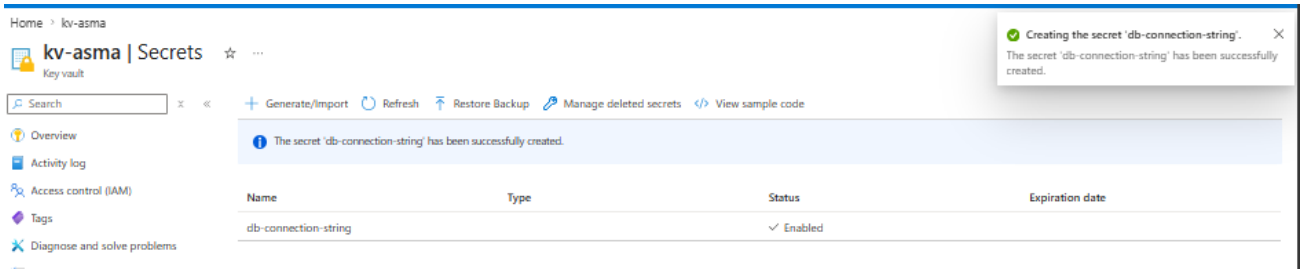
6. Security



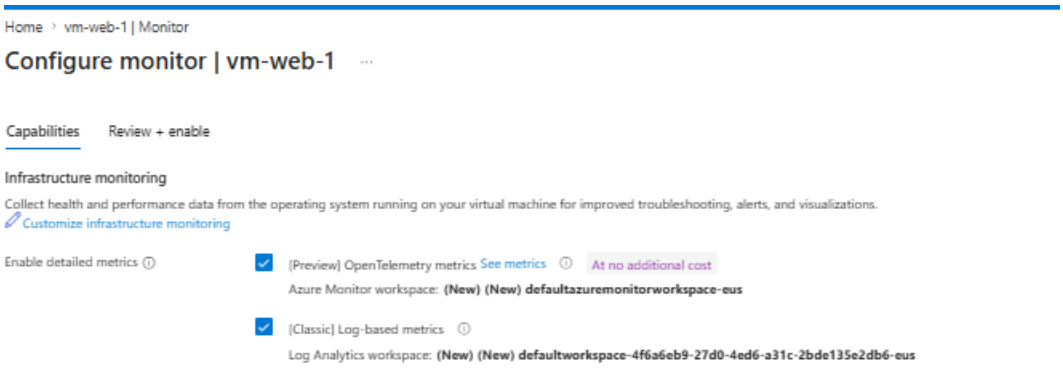


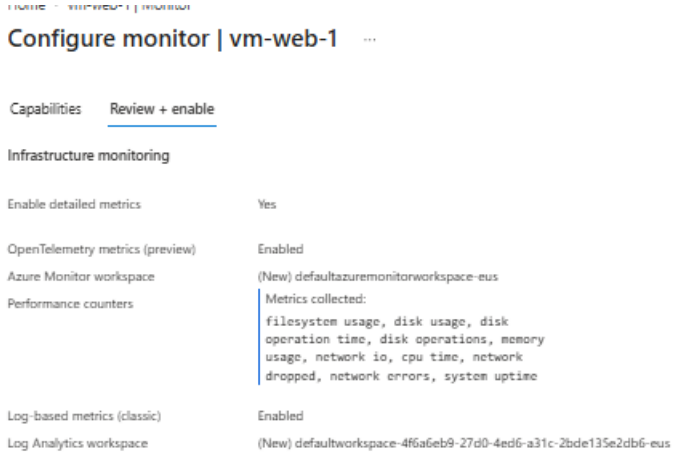


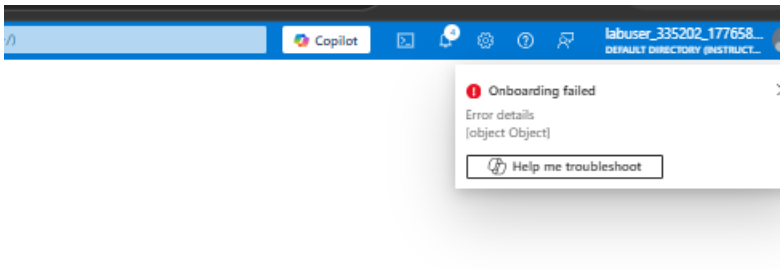


Azure Key Vault was used to securely store sensitive data such as connection strings.

7. Monitoring & Logging







Create an alert rule

Scope | Condition | Actions | Details | Tags | Review + create

Product details

Metric alert rule	Total pricing
1 Condition	0.10 USD/month
Terms of use | Privacy statement	Pricing

Scope

Scope level	Subscription
Resource	PAYG-Sandboxes > rg_sb_eastus_335202_1_177658078478 > vm-web-1

Condition

Signal name	Percentage CPU
Operator	Greater than
Aggregation type	Average
Threshold value	70
Lookback period	5 minutes
Check every	1 minute

Details

Project details

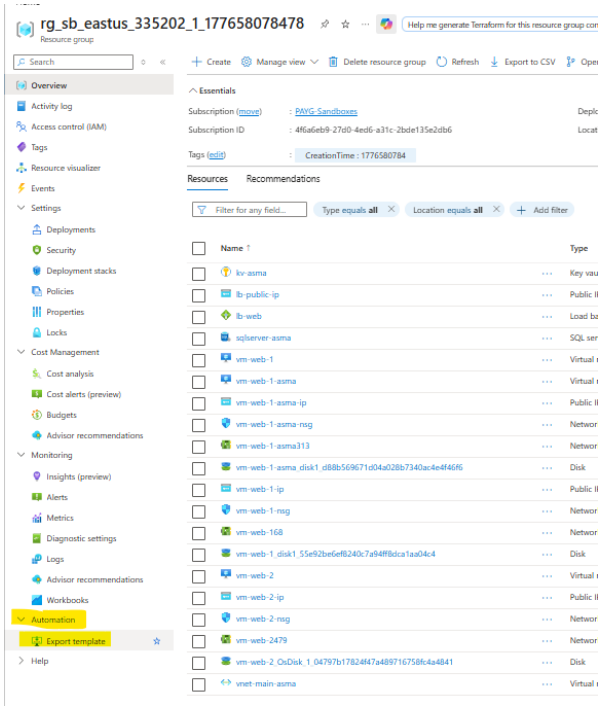
Subscription	PAYG-Sandboxes
Resource group	rg_sb_eastus_335202_1_177658078478
Region	global

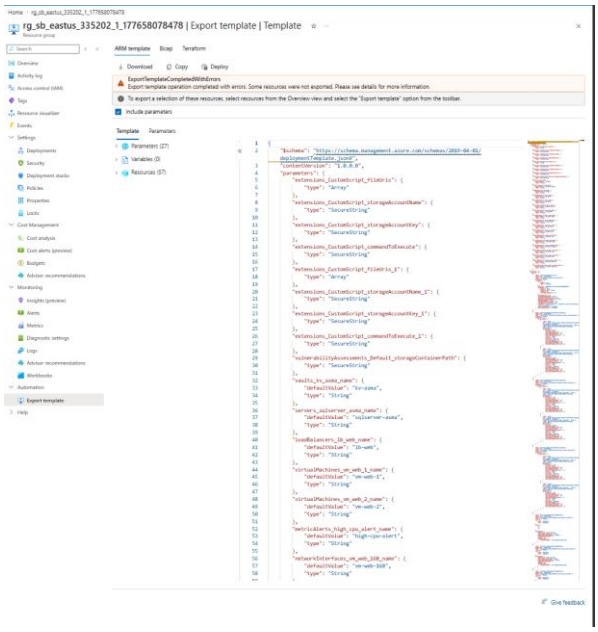
Alert rule details

Alert rule name	high-cpu-alert
Alert rule description	
Severity	3 - Informational
Enable upon creation	☑
Automatically resolve alerts	☑

Azure Monitor and Alerts were configured to monitor system performance.

8. Automation (Mandatory)



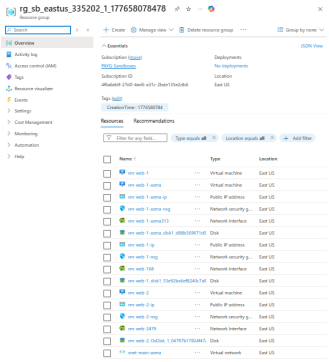


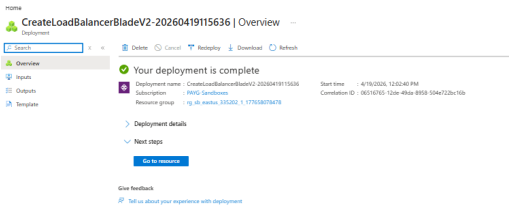
ARM Template was exported to enable infrastructure automation and reproducibility.

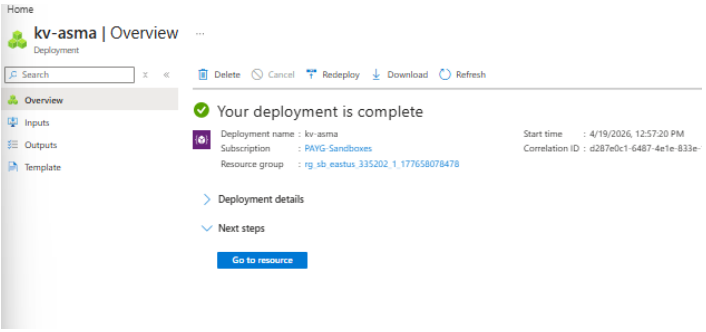
1. Architecture Diagram

User
↓
Load Balancer
↓
VM1 VM2
↓
SQL Database
↑
Key Vault

Deployment Proof







The infrastructure was successfully deployed using Azure.

Due to lab time expiration, resources are no longer accessible, but the exported ARM template confirms the deployment.

Screenshots of the deployed resources (VMs, Load Balancer, SQL Database, and Key Vault) were taken during the lab to verify successful deployment.